

BSR Quality Checker - Product Overview



AI-Powered Building Safety Compliance Platform

Tagline: Automated compliance analysis in minutes, not days. AI-powered intelligence that replaces weeks of manual consultancy work.

The Problem We Solve

Traditional Consultancy Approach (Status Quo)

Building safety consultants currently spend **5-20 hours per project** manually:

- Reading through hundreds of pages of Gateway 2 submission documents
- Cross-referencing 55+ Building Safety Regulator (BSR) requirements
- Creating compliance matrices mapping requirements → evidence → page numbers
- Identifying gaps and missing information
- Writing client reports and action plans
- Billing £500-1,000+ per compliance matrix (taking days to produce)

Pain Points:

- **✗** Manual document review is slow and error-prone
- **✗** Human reviewers miss requirements or inconsistencies
- **✗** Compliance matrices take 4-8 hours to build manually
- **✗** Clients wait days for gap analysis feedback
- **✗** High cost per project limits market accessibility

-  No standardization - quality varies by consultant

Our AI-Powered Solution

BSR Quality Checker automates the entire compliance review process using AI:

-  Upload documents → AI analysis complete in **2-5 minutes**
 -  Automated requirement checking across 55 BSR criteria
 -  AI-generated compliance matrices (worth £500-1,000) in **seconds**
 -  Instant gap analysis with page-level evidence linking
 -  Three submission-ready documents generated automatically
 -  Consistent, repeatable quality every time
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What Is BSR Quality Checker?

BSR Quality Checker is an **AI-powered SaaS platform** that automates Building Safety Regulator (BSR) Gateway 2 compliance assessments for UK high-rise residential buildings (18m+).

Core Product: AI Compliance Analysis Engine

Input:

- Fire Strategy documents
- Structural calculations
- Architectural drawings
- Building specifications
- Design & Access Statements
- Any Gateway 2 submission documents

AI Processing:

1. **Document Intelligence:** AI extracts text, requirements, evidence, and context from uploaded PDFs

2. **Compliance Mapping:** AI checks each document against 55 proprietary BSR compliance rules
3. **Evidence Linking:** AI maps every requirement to specific document + page number with quotes
4. **Gap Analysis:** AI identifies missing information, non-compliant areas, and ambiguities
5. **Action Planning:** AI generates prioritized action items with owner assignments

Output (Generated in 2-5 minutes):

1. **Client Gap Analysis PDF** - What information the client needs to provide
 2. **Consultant Action Plan PDF** - Internal working document with prioritized issues
 3. **Compliance Matrix Excel** - Submission-ready traceability matrix with color-coded status
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Product Architecture & Technology

Five-Stage Product Lifecycle (Inspired by Patent Prosecution Workflow)

Stage 1: Discovery & Gap Analysis LIVE

- Quick document upload and instant AI assessment
- Identifies missing requirements and compliance gaps
- Generates gap analysis reports for client follow-up
- **Current Phase:** Fully operational, production-ready

Stage 2: Submission Drafting PLANNED (6-9 months)

- AI-assisted section generation for BSR submissions
- Template library for common building types
- Auto-fill capabilities based on extracted data
- **Goal:** Reduce drafting time from weeks to hours

Stage 3: Evidence Linking (Compliance Matrix Generator) LIVE

- Auto-generates professional Excel compliance matrices
- Maps requirements → evidence → document → page number
- Color-coded status (Met/Partial/Not Met)
- **Value:** £500-1,000 per matrix, generated in seconds

Stage 4: Review Workflow 🧠 PLANNED (9-12 months)

- Internal review routing for multi-specialist sign-offs
- Specialist assignment and collaboration tools
- Pre-submission quality checks
- **Goal:** Streamline multi-party review process

Stage 5: Post-Submission Management 🧠 PLANNED (9-12 months)

- RFI (Request for Information) response generator
- Submission status tracking and audit trail
- Regulator communication management
- **Goal:** Handle post-submission workflow efficiently

Technical Stack

Frontend:

- React + TypeScript
- Vite build system
- Tailwind CSS for styling
- Client-side PDF preview
- Real-time progress tracking

Backend:

- Node.js + Express
- TypeScript for type safety
- Prisma ORM (PostgreSQL production, SQLite local dev)
- Puppeteer for PDF generation
- ExcelJS for compliance matrix export

AI/ML Layer:

- Anthropic Claude Sonnet 4.5 for document analysis
- Custom prompt engineering for compliance assessment
- 55 proprietary deterministic rules engine
- Hybrid approach: Rules + AI enrichment

Infrastructure:

- Hosted on Railway (automatic deployment from GitHub)
 - PostgreSQL database
 - File storage for document processing
 - API-first architecture
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Key Features

1. Quick Assess (Main User Flow)

Purpose: Run instant compliance checks on any building project

Workflow:

1. Upload documents (PDFs: fire strategy, drawings, specs, etc.)
2. AI processes documents and runs 55 compliance checks (2-5 min)
3. View results in intuitive dashboard with color-coded status
4. Download 3 AI-generated documents:
 - Client Gap Analysis PDF
 - Consultant Action Plan PDF
 - Compliance Matrix Excel
5. Save assessment to client record for future reference

Use Cases:

- Initial project feasibility check
- Pre-submission review

- Client onboarding gap analysis
- Proposal generation for consultancy bids

2. Client Management System

Purpose: Organize projects by client for consultancy workflow

Features:

- Create clients with contact information
- Link multiple packs (projects) to each client
- Version history for iterative assessments
- Download reports for any saved assessment
- Track project status and progress

Use Cases:

- Agency/consultancy managing multiple clients
- Tracking repeat clients across projects
- Organizing by building developer or architect firm

3. AI-Generated Compliance Matrix (Stage 3)

Purpose: Auto-generate submission-ready traceability matrices

What It Produces:

- Professional Excel spreadsheet with:
 - **Summary section:** Project info, compliance statistics (Met/Partial/Not Met percentages)
 - **Data table:** ID | Requirement | Category | Status | Priority | Evidence Document | Page | Action | Owner | Notes
 - **Color coding:** Green (Met), Yellow (Partial), Red (Not Met)
 - **Formatting:** Frozen headers, auto-filters, borders, alternating rows
 - **Evidence linking:** Every requirement mapped to specific document + page number

Value Proposition:

- Traditional consultancies: £500-1,000 per matrix, 4-8 hours manual work
- BSR Quality Checker: £0 (included), generated in seconds by AI
- **ROI:** Instant, repeatable, consistent quality

4. 55-Point Compliance Rules Engine

Categories Covered:

- Fire Safety Strategy requirements
- Structural compliance
- MEP (Mechanical, Electrical, Plumbing) systems
- Architectural design standards
- Design & Access Statement completeness
- Submission formatting and documentation
- Environmental considerations
- Accessibility compliance

Rules Examples:

- "Fire strategy must specify evacuation strategy (stay put/simultaneous)"
- "Building height must be stated explicitly in meters"
- "Compartmentation details required for residential buildings 18m+"
- "Structural calculations must reference Eurocodes"
- "External wall system materials must be explicitly identified"

5. AI-Powered Document Analysis

What the AI Does:

- Extracts building type, height, location, storeys from context
- Identifies whether building is HRB (High-Rise Building) or London-regulated
- Reads fire strategies and maps to compliance requirements
- Cross-references evidence across multiple documents
- Identifies ambiguities and areas requiring clarification
- Generates human-readable reasoning for each assessment

- Suggests specific actions with owner assignments
- Estimates effort levels (S/M/L) for remediation

AI Advantages Over Manual Review:

- **Speed:** 100x faster than human reading
 - **Consistency:** Same criteria applied every time
 - **Coverage:** Never misses a requirement
 - **Evidence linking:** Automatically finds page numbers and quotes
 - **Reasoning:** Explains WHY something fails or passes
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User Personas

Primary: Building Safety Consultants

Profile:

- Fire safety engineers, structural engineers, building control approvers
- Managing 5-50 Gateway 2 projects simultaneously
- Billing £80-150/hour for compliance reviews
- Spending 5-20 hours per project on manual document review

Pain Points:

- Drowning in document review work
- Missing requirements during manual checks
- Time-consuming matrix building
- Clients demanding faster turnaround

How We Help:

- Run assessment in 5 minutes vs 5 hours
- Never miss a requirement (AI + rules engine)
- Auto-generate matrices worth £500-1,000
- Instant client gap reports for faster follow-up

Secondary: Architects & Developers

Profile:

- Architectural practices submitting Gateway 2 applications
- Building developers coordinating multi-specialist teams
- Need compliance checks before engaging consultants

Pain Points:

- Don't know if submission is complete before paying consultants
- Expensive consultancy fees for basic gap checks
- Waiting weeks for consultant availability

How We Help:

- Self-serve compliance checks before consultant engagement
- Identify gaps early to prepare complete submissions
- Reduce rework and resubmission costs

Tertiary: Building Control Bodies

Profile:

- Approved Inspectors and Local Authority Building Control
- Reviewing Gateway 2 submissions for compliance
- Need standardized checklists and audit trails

Pain Points:

- Manual review is slow and inconsistent
- Hard to track compliance status across projects
- No audit trail for decisions

How We Help:

- Standardized compliance framework
- Documented evidence links for audit trail
- Fast initial review before detailed assessment

Business Model & Pricing

Current Status: Pre-Revenue / MVP

Future Pricing Models (Under Consideration):

Option 1: Per-Assessment Pricing

- £99 per Quick Assess (single-use)
- £49 for repeat assessments on same project
- No subscription, pay-as-you-go

Option 2: Subscription Tiers

- **Starter:** £199/month - 5 assessments/month
- **Professional:** £499/month - 20 assessments/month
- **Agency:** £999/month - Unlimited assessments + team features

Option 3: Freemium + Premium Features

- Free: Basic gap analysis (limited features)
- Pro: £299/month - Full compliance matrix, unlimited assessments
- Enterprise: Custom pricing - API access, white-label, integrations

Revenue Comparison:

- Traditional consultant: £500-1,000 per compliance matrix (one-time)
- BSR Quality Checker: £99-299/month (recurring) for unlimited matrices

Competitive Differentiation

vs. Traditional Consultancy Services

Traditional Consultancy	BSR Quality Checker
 5-20 hours per project	 2-5 minutes
 £500-1,000 per matrix	 £99-299/month unlimited
 Manual document review	 AI-powered automation
 Quality varies by consultant	 Consistent, repeatable
 Requires human availability	 24/7 self-service
 Email delivery (days)	 Instant download

vs. Checklist/Template Tools

Competitors: Generic building compliance checklists, Excel templates

Our Advantages:

-  They require manual input →  We auto-extract from documents
-  Static checklists →  AI-powered intelligence
-  No evidence linking →  Automatic requirement → page mapping
-  No reasoning →  AI explains WHY and HOW to fix
-  One-size-fits-all →  Adapts to building type (HRB, London, etc.)

vs. Generic Document AI Tools

Competitors: ChatGPT, general-purpose document analyzers

Our Advantages:

-  No BSR domain expertise →  55 proprietary compliance rules
-  Generic outputs →  Submission-ready matrices and reports
-  No structured workflow →  End-to-end consultancy process
-  Requires prompt engineering →  One-click assessment
-  No compliance framework →  Built for UK Building Safety Act

Market Opportunity

Total Addressable Market (TAM)

UK Gateway 2 Submissions (Annual):

- ~5,000-10,000 high-rise buildings requiring Gateway 2 submissions
- Each building requires compliance review (£500-2,000 per review)
- Total market: £2.5M-20M annually for compliance analysis alone

Serviceable Market:

- Building safety consultancies: ~500 firms in UK
- Architectural practices: ~2,000 firms handling high-rise projects
- Building control bodies: ~100 organizations

Early Adopter Segments:

- Small consultancies (1-5 engineers) lacking capacity
- Mid-size firms (10-50 staff) seeking efficiency gains
- Architects doing pre-submission checks

Growth Trajectory

Phase 1 (Months 0-6): Stage 3 Launch - Compliance Matrix Focus

- Target: 20-50 paying consultancy customers
- Revenue: £5,000-15,000 MRR
- Focus: Perfect compliance matrix product-market fit

Phase 2 (Months 6-12): Stage 2 Launch - Submission Drafting

- Target: 100-200 customers
- Revenue: £30,000-60,000 MRR
- Focus: Expand to full submission workflow

Phase 3 (Months 12-24): Stages 4-5 - Review & Post-Submission

- Target: 500-1,000 customers
 - Revenue: £150,000-300,000 MRR
 - Focus: Enterprise deals with large consultancies
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Product Roadmap

Completed (Current State)

Stage 1: Discovery & Gap Analysis

- Quick Assess workflow (upload → analyze → download reports)
- 55-point compliance rules engine
- AI-powered document analysis with Claude Sonnet 4.5
- Client management system
- Version history and saved assessments

Stage 3: Compliance Matrix Generator

- AI-generated Excel compliance matrices
- Color-coded status (Met/Partial/Not Met)
- Evidence linking (requirement → document → page)
- Professional formatting (frozen headers, filters, borders)
- Summary statistics and compliance rate calculation

Infrastructure:

- Production deployment on Railway
- PostgreSQL database
- PDF generation pipeline
- Excel export with ExcelJS
- TypeScript full-stack application

In Progress

UI/UX Improvements:

- Simplified results view (single-screen clarity)
- AI branding throughout product (differentiate from consultancies)
- Download button reliability fixes

Backend Optimizations:

- Payload size handling (10MB limit for large assessments)
- Database schema for client/pack relationships
- Caching for faster repeat assessments

Planned (Next 6-12 Months)

Stage 2: Submission Drafting (Q2-Q3 2026)

- AI-assisted section generation for Gateway 2 submissions
- Template library for common building types
- Auto-fill based on extracted data
- Collaborative editing for multi-specialist teams

Stage 4: Review Workflow (Q3-Q4 2026)

- Specialist assignment and routing
- Internal review sign-offs
- Pre-submission quality checks
- Team collaboration features

Stage 5: Post-Submission Management (Q4 2026)

- RFI response generator
- Status tracking and notifications
- Regulator communication log
- Full audit trail

Enterprise Features:

- API access for integration with existing tools

- White-label for large consultancies
- Custom compliance frameworks
- SSO and team management

Technical Deep Dive

AI Compliance Assessment Workflow

Step 1: Document Upload & Processing

User uploads PDFs → Backend stores files → AI extracts text + metadata

- Accepts multiple PDFs (fire strategy, drawings, specs, etc.)
- Extracts building context (type, height, location, storeys)
- Determines if High-Rise Building (HRB) or London-regulated

Step 2: Deterministic Rules Engine

55 if-then rules → Check each requirement → Output: meets/partial/does_not_meet

- Example rules:
 - IF (building height $\geq$ 18m) THEN (fire strategy required)
 - IF (fire strategy present) THEN (evacuation strategy must be specified)
 - IF (London location) THEN (additional London Building Act requirements apply)

Step 3: AI Enrichment (Claude Sonnet 4.5)

For each requirement → AI reads documents → Extracts evidence + reasoning

- AI finds relevant sections in documents
- Links to specific page numbers
- Extracts supporting quotes

- Identifies gaps (what's missing)
- Generates human-readable reasoning
- Suggests remediation actions

Step 4: Triage & Prioritization

AI categorizes issues → Assigns urgency → Determines owner → Estimates effort

- Critical blockers (must fix before submission)
- High priority (significant compliance risk)
- Medium priority (should address)
- Low priority (nice to have)

Step 5: Report Generation

Assessment data → Generate PDFs (Puppeteer) + Excel (ExcelJS) → Download

- Client Gap Analysis: What client needs to provide
- Consultant Action Plan: Internal working document
- Compliance Matrix: Submission-ready traceability

Data Model

Core Entities:

- **Client:** Name, company, contact info, notes
- **Pack:** Project/submission package linked to client
- **PackVersion:** Version history for iterative assessments
- **Document:** Uploaded PDFs with metadata
- **PackTask:** Checklist items for tracking progress
- **AssessmentResult:** Compliance status for each requirement

Key Relationships:

Client (1) → (many) Packs
Pack (1) → (many) PackVersions

PackVersion (1) → (many) Documents
PackVersion (1) → (many) AssessmentResults
Pack (1) → (many) PackTasks

API Endpoints

Assessment:

- `POST /api/assess` - Run new assessment
- `POST /api/assess/save` - Save assessment to client

Export:

- `POST /api/packs/:packId/versions/:versionId/client-gap-analysis/download`
- `POST /api/packs/:packId/versions/:versionId/consultant-action-plan/download`
- `POST /api/packs/:packId/versions/:versionId/compliance-matrix/excel`
- `GET /api/packs/:packId/versions/:versionId/saved-assessment/compliance-matrix/excel`

Clients:

- `GET /api/clients` - List all clients
- `GET /api/clients/:id` - Get client with packs
- `POST /api/clients` - Create new client
- `DELETE /api/clients/:id` - Delete client

Packs:

- `GET /api/packs/:id` - Get pack details
- `POST /api/packs` - Create new pack
- `DELETE /api/packs/:id` - Delete pack

Success Metrics & KPIs

Product Metrics

- **Time to assessment:** Target <5 minutes (currently 2-5 min) 
- **Accuracy rate:** % of AI assessments matching human expert review (target 90%+)
- **Evidence linking success:** % of requirements with valid document + page references
- **Report generation success:** % of assessments producing all 3 documents without errors

Business Metrics

- **Monthly Active Users (MAU):** Target 20-50 in first 6 months
- **Assessments per user per month:** Target 5-10 (indicates value/adoption)
- **Conversion rate:** Free → Paid (if freemium model)
- **Monthly Recurring Revenue (MRR):** Target £5K-15K in first 6 months
- **Churn rate:** Target <5% monthly
- **Net Promoter Score (NPS):** Target 50+ (consultants would recommend)

Efficiency Metrics (vs. Traditional Consultancy)

- **Time savings:** 95%+ reduction (5 hours → 5 minutes)
- **Cost savings:** 80%+ reduction (£500-1,000 → £99-299/month unlimited)
- **Compliance matrix generation:** 100x faster (4-8 hours → 30 seconds)

Risk Factors & Mitigations

Technical Risks

AI Accuracy & Hallucinations

- **Risk:** AI misses requirements or generates incorrect assessments
- **Mitigation:** Hybrid approach (deterministic rules + AI enrichment), human review workflows in Stage 4, continuous validation against expert reviews

Document Format Variability

- **Risk:** PDFs vary widely (scanned images, tables, complex layouts)
- **Mitigation:** Robust text extraction, support for OCR, clear guidance on acceptable formats

Scalability & Performance

- **Risk:** AI processing slow for large document sets (100+ page PDFs)
- **Mitigation:** Async processing, progress tracking, caching, chunking strategies

Market Risks

Regulatory Changes

- **Risk:** BSR requirements change, invalidating rules engine
- **Mitigation:** Modular rules engine design, rapid update capabilities, compliance versioning

Consultant Resistance

- **Risk:** Consultants view as threat to their business
- **Mitigation:** Position as productivity tool, not replacement; target junior engineers and small firms first; emphasize time savings for higher-value work

Slow Adoption

- **Risk:** Market not ready for AI-powered compliance tools
- **Mitigation:** Pilot programs with early adopters, case studies showing ROI, freemium to reduce barrier to entry

Legal/Liability Risks

Professional Indemnity

- **Risk:** Who is liable if AI assessment is wrong and causes submission failure?
- **Mitigation:** Clear disclaimers ("All findings should be reviewed by qualified professionals"), position as decision-support tool, not final authority

Data Privacy & Security

- **Risk:** Handling sensitive building plans and client data
 - **Mitigation:** SOC 2 compliance pathway, encryption at rest/in transit, GDPR compliance, clear data retention policies
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Why Now? Market Timing

Regulatory Drivers:

- UK Building Safety Act 2022 created Gateway 2 regime
- Thousands of high-rise buildings now require compliance assessments
- Consultancy bottleneck: Not enough qualified reviewers to meet demand

Technology Enablers:

- Claude Sonnet 4.5 (Jan 2025 release) offers unprecedented document understanding
- AI can now reliably extract requirements from complex technical documents
- PDF processing and AI integration mature enough for production use

Market Conditions:

- Building safety consultancies overwhelmed with demand post-Grenfell
 - Clients demanding faster turnaround and lower costs
 - Industry ready for automation (proven by early positive feedback)
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Conclusion: The Vision

BSR Quality Checker is building the operating system for building safety compliance.

We're not just automating compliance checks—we're creating an **end-to-end platform** that handles the entire Gateway 2 submission lifecycle:

-  **Stage 1:** Discover gaps (live)

- 🛠️ **Stage 2:** Draft submissions (planned)
- ✅ **Stage 3:** Generate compliance matrices (live)
- 🧑‍🔬 **Stage 4:** Coordinate reviews (planned)
- 🧑‍🔬 **Stage 5:** Manage post-submission (planned)

Our competitive moat:

1. **Domain expertise:** 55 proprietary BSR compliance rules
2. **AI engineering:** Custom prompts optimized for building safety
3. **Workflow design:** Consultancy-focused UX built for professionals
4. **Data network effects:** More assessments → better AI training → better product
5. **First-mover advantage:** No direct AI-powered competitors in BSR compliance

The opportunity:

- Replace £500-1,000 manual compliance matrices with AI-generated versions (seconds, not hours)
- Expand to full submission drafting and review workflow
- Become the de facto standard for UK building safety compliance
- International expansion (Australia, Canada, EU have similar regimes)

The mission: Make building safety compliance faster, cheaper, and more reliable through AI automation—helping ensure no building slips through the cracks due to manual review errors or capacity constraints.

Contact & Next Steps

For Investors:

- See our competitive analysis and go-to-market strategy
- Review financial projections and unit economics
- Schedule product demo and customer interviews

For Early Customers:

- Try Quick Assess with your next Gateway 2 project

- See compliance matrix generation in action
- Provide feedback to shape the product roadmap

For Partners:

- Integration opportunities with existing building safety tools
- White-label for large consultancies
- API access for custom workflows

Last Updated: March 2026 Version: 2.0 (Post-Stage 3 Launch) 🚢 AI-Powered Building Safety Compliance